

Set operation:

Its the collection of unique values.

Values are written in curly brackets {}

It doesn't allow duplicates

order might change automatically

Why do we use set?

only unique values.

fast checking.

mathematical operation like Union & Intersection.

```
In [1]: a = {1,2,3,4,5}
        b = {4,5,6,7,8}
        c = {8,9,10}
```

Union |(Pipe symbol) : combines all values from both sets.

```
In [2]: a.union(b)
```

```
Out[2]: {1, 2, 3, 4, 5, 6, 7, 8}
```

```
In [3]: a.union(a,b)
```

```
Out[3]: {1, 2, 3, 4, 5, 6, 7, 8}
```

```
In [4]: a.union(c)
```

```
Out[4]: {1, 2, 3, 4, 5, 8, 9, 10}
```

```
In [5]: a.union(a,c)
```

```
Out[5]: {1, 2, 3, 4, 5, 8, 9, 10}
```

```
In [6]: a|b Here | Pipe symbol represents 'Union'
```

```
Out[6]: {1, 2, 3, 4, 5, 6, 7, 8}
```

```
In [7]: print(a)
        print(b)
        print(c)
```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

Union intersection (&): it finds common values in both sets OR Only the values present in both sets.

```
In [8]: a.intersection(b)
```

```
Out[8]: {4, 5}
```

```
In [9]: a.intersection(c) # Here there is no common element in both set, so it displays empty set.
```

```
Out[9]: set()
```

```
In [10]: a.intersection(b)
```

```
Out[10]: {4, 5}
```

```
In [11]: a & b
```

```
Out[11]: {4, 5}
```

```
In [12]: a & c
```

```
Out[12]: set()
```

```
In [14]: a & b & c
```

```
Out[14]: set()
```

Difference (-): It removes common value from the set

```
In [15]: print(a)
        print(b)
        print(c)
```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

```
In [16]: a.difference(b) # it won't display common element
```

```
Out[16]: {1, 2, 3}
```

```
In [17]: a.difference(c)
```

```
Out[17]: {1, 2, 3, 4, 5}
```

```
In [18]: b.difference(a)
```

```
Out[18]: {6, 7, 8}
```

Symtenric difference (^): It elimenates common values from both the sets and displays remaining element.

```
In [19]: print(a)
         print(b)
         print(c)
```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

```
In [20]: a.symmetric_difference(b)
```

```
Out[20]: {1, 2, 3, 6, 7, 8}
```

```
In [21]: a.symmetric_difference(c)
```

```
Out[21]: {1, 2, 3, 4, 5, 8, 9, 10}
```

```
In [22]: b.symmetric_difference(a)
```

```
Out[22]: {1, 2, 3, 6, 7, 8}
```

```
In [23]: c.symmetric_difference(b)
```

```
Out[23]: {4, 5, 6, 7, 9, 10}
```

```
In [24]: a^b^c
```

```
Out[24]: {1, 2, 3, 6, 7, 9, 10}
```

```
In [ ]: # Symentric difference update: it displays uncommon element
        # keeps only the values that are different in both sets.
        # And it updates the original set directly.

        # Renoves:

        # Common values
        # Keeps uncommon values
```

```
In [25]: print(a)
         print(b)
         print(c)
```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

```
In [30]: a.symmetric_difference_update(b)
```

```
In [31]: print(a)
         print(b)
         print(c)
```

```
{1, 2, 3, 6, 7, 8}  
{4, 5, 6, 7, 8}  
{8, 9, 10}
```

```
In [34]: a.symmetric_difference_update(b)
```

Intersection: It finds common values from both the sets.

```
In [42]: a1 = {1, 2, 3, 4, 5}  
b1 = {4, 5, 6, 7, 8}  
c1 = {8, 9, 10}
```

```
In [35]: a1.intersection(b1)
```

```
Out[35]: {4, 5}
```

```
In [36]: a1.intersection(c1) # here in c1 there are no common values so it returns empty set.
```

```
Out[36]: set()
```

```
In [37]: b1.intersection(c1)
```

```
Out[37]: {8}
```

```
In [38]: a1.intersection(c1) # here in c1 there are no common values so it returns empty set.
```

```
Out[38]: set()
```

```
In [ ]: # Update: It will update the sets a1 & b1 together and display all the elements together.
```

```
In [18]: a1 = {1, 2, 3, 4, 5}  
b1 = {4, 5, 6, 7, 8}  
c1 = {8, 9, 10}
```

```
In [19]: print(a1)  
print(b1)  
print(c1)
```

```
{1, 2, 3, 4, 5}  
{4, 5, 6, 7, 8}  
{8, 9, 10}
```

```
In [32]: a1.update(b1)  
a1
```

```
Out[32]: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
```

```
In [31]: b1.update(c1)  
b1
```

```
Out[31]: {4, 5, 6, 7, 8, 9, 10}
```

```
In [30]: a1.update(c1)  
a1
```

```
Out[30]: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
```

```
In [34]: a1.intersection_update(b1) # Here it removes the common element  
a1
```

Out[34]: {4, 5, 6, 7, 8, 9, 10}

```
In [36]: c1.intersection_update(b1)
c1
```

Out[36]: {8, 9, 10}

```
In [ ]: # Superset: A set that contains all elements of another set is called a superset.

# Subset: Smaller set inside another set

# Disjoint: Two sets having no common elements are called disjoint sets.
```

```
In [43]: A1 = {1,2,3,4,5,6,7,8,9}
B1 = {3,4,5,6,7,8}
C1 = {10,20,30,40}
```

```
In [44]: print(A1)
print(B1)
print(C1)
```

```
{1, 2, 3, 4, 5, 6, 7, 8, 9}
{3, 4, 5, 6, 7, 8}
{40, 10, 20, 30}
```

```
In [45]: B1.issubset(A1) # Here A1 & B1 have similar numbers in set, so it shows True
```

Out[45]: True

```
In [47]: A1.issuperset(B1)
```

Out[47]: True

```
In [48]: A1.issuperset(C1) # Here A1 & C1 different set values, so its shows false.
```

Out[48]: False

```
In [ ]:
```